Synthesis of Long-Range Dependent Symbol Sequences

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1 Annotated Bibliography

This document starts the literature foundation for S5.jl, a package for generating long-range dependent non-numerical symbol sequences. The present version is an annotated bibliography rather than a full paper draft. Reference PDFs should be placed in this directory when available. Items not yet downloaded are listed in `DOWNLOAD.mn`.

1.1 Long-Range Dependence and Self-Similarity

Leland, Taqqu, Willinger, and Wilson (1994). *On the self-similar nature of Ethernet traffic.* This paper is one of the central empirical starting points for modern LRD network-traffic modelling. It showed that packet traffic could retain burstiness across aggregation scales, motivating models beyond Poisson and short-memory Markov assumptions. For S5.jl it supplies the broader methodological lesson: synthetic test data should reproduce large-scale structure, not just local statistics.

Pipiras and Taqqu (2017). *Long-Range Dependence and Self-Similarity.* This monograph provides the mathematical background tying together covariance decay, spectral poles, self-similarity, and the Hurst parameter. It is the main reference for the parameter relationships used throughout the package, including the links between spectral exponent, ACF decay exponent, and Hurst parameter.

Beran (1994). *Statistics for Long-Memory Processes.* Beran's text is a core statistical reference for inference and modelling in long-memory time series. Although S5.jl is not an estimator package, the book helps frame why estimator benchmarking needs synthetic sequences with known parameters and known violations of short-memory assumptions.

Taqqu, Teverovsky, and Willinger (1995). *Estimators for long-range dependence: an empirical study.* This empirical comparison of LRD estimators motivates the separation between S5.jl and a future estimator package. The paper demonstrates that estimator performance depends heavily on model class, sample size, and diagnostics, which is why S5.jl focuses first on controllable generators.

1.2 Numerical LRD Synthesis

Paxson (1997). *Fast, approximate synthesis of fractional Gaussian noise for generating self-similar network traffic.* This is the direct ancestor of S5.jl's PB1 method. It gives a practical spectral construction of fGn, trading exact covariance matching for speed. In S5.jl the numerical fGn sequence is mapped to symbols by rank binning, preserving a simple path from target Hurst parameter to categorical output.

Dieker (2004). *Simulation of fractional Brownian motion.* Dieker's thesis surveys exact and approximate fBm/fGn simulation methods, including circulant embedding. It provides the natural next reference for improving PB1 if exact covariance control becomes more important than speed.

Roughan, Veitch, and Abry (2001). *Real-time estimation of the parameters of long-range dependence.* This work is important for the wavelet perspective behind the planned PB3 generator. Wavelet methods are attractive because they align naturally with multi-scale structure and may provide a bridge from LRD synthesis to local state-machine control.

1.3 Point Processes and Heavy-Tailed Models

Garrett and Willinger (1994). *Analysis, modeling, and generation of self-similar VBR video traffic.* This paper is a classic example of generating self-similar traffic from heavy-tailed source behaviour. It motivates MB2's heavy-tailed regime-sojourn construction, where long periods in a regime create large-scale dependence while the within-regime Markov chain controls local symbol structure.

Lowen and Teich (1995). *Estimation and simulation of fractal stochastic point processes.* Lowen and Teich provide the point-process foundation for fractal renewal and shot-noise constructions. Their work informs MB3's view of each symbol stream as an independent heavy-tailed arrival process.

Ryu and Lowen (1998). *Point process models for self-similar network traffic, with applications.* This paper develops point-process models that connect heavy-tailed mechanisms to self-similar traffic. It is a key bridge between network-traffic synthesis and the symbolic FSS construction used in S5.jl.

Roughan, Yates, and Veitch (1999). *The mystery of the missing scales: pitfalls in the use of fractal renewal processes to simulate LRD processes.* This paper is a warning label for naive heavy-tailed simulation. It shows that FRP constructions can fail to express LRD over the intended scale range. S5.jl's validation scripts and TODO list retain this as an explicit design concern for FSS and On/Off methods.

1.4 Symbolic and Categorical Models

Gal, Chen, and Ghahramani (2015). *Latent Gaussian processes for distribution estimation of multivariate categorical data.* This work motivates PB2. The central idea is to represent categorical outcomes through latent Gaussian variables and an argmax transformation. S5.jl adapts that idea to long-memory latent fGn streams and adds finite-sample offset calibration for marginal control.

Kumar, Raghu, Sarlos, and Tomkins (2017). *Linear additive Markov processes.* This paper motivates MB1. LAMP models make transition probabilities additive in past observations, allowing long-range memory to be encoded through a decay kernel over history. S5.jl implements a finite-depth version with a marginal innovation term.

Singh, Greenberg, and Klakow (2016). *The custom decay language model for long range dependencies.* This language-modelling paper is closely related to LAMP in spirit: it uses a custom decay over historical context to model long-range dependencies in text. It supports the idea that symbolic sequences need memory mechanisms distinct from ordinary short-context Markov models.

Carpio and Daley (2007). *Long-range dependence of Markov chains in discrete time on countable state space.* This paper is important because it frames LRD for Markov chains through counting and recurrence-time behaviour. It helps justify the use of count-process and return-time definitions for non-numerical sequences where ordinary numerical ACF methods are not directly applicable.

1.5 Machine Learning Motivation

Belletti, Beutel, Jain, and Chi (2018). *Factorized recurrent neural architectures for longer range dependence.* This paper motivates the project from the machine-learning side by showing that common recurrent architectures struggle with long-range dependence. It helps explain why synthetic NN-LRD data is useful for stress-testing learning algorithms.

Belletti, Chen, and Chi (2019). *Quantifying long range dependence in language and user behavior to improve RNNs.* This paper is one of the direct motivations for studying LRD in language and human behavioural sequences. It connects empirical LRD measurement to model design, reinforcing S5.jl's role as a controllable data source for later estimator and learning experiments.